

Teleoperation for Bimanual Dexterous Manipulation via Multi-Demonstration Residual Learning

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Abstract—Reinforcement learning guided by human demonstrations can produce high-fidelity bimanual dexterous manipulation in simulation, but existing methods fit a separate residual policy to each reference trajectory. Such a policy memorises one kinematic path and is brittle when the input deviates from it, which is unavoidable under live teleoperation. We train a single shared residual policy across a *set* of demonstrations that vary in starting configuration and approach direction, on top of a frozen hand-motion imitator, so that the policy learns the manipulation skill rather than a specific trajectory. To collect these demonstrations and to drive the policy at run time we build a bimanual capture pipeline that fuses Apple Vision Pro hand tracking with OptiTrack object poses through a cross-system Umeyama calibration and streams them into Isaac Gym at 60 Hz. On two long-horizon bimanual tasks — capping an alcohol burner and flipping a brush — the residual policy is the only method evaluated that completes the full sequence offline, seating the cap in 7/10 held-out demonstrations where an optimization-based retargeter seats it in 0/10 despite transporting it perfectly. Under live teleoperation the same policy, without retraining, completes the capping task on trajectories never seen during training.

I. INTRODUCTION

Dexterous bimanual manipulation — tasks requiring two hands to coordinate precisely around a shared object — remains one of the most challenging problems in robotics. Contact-rich interactions demand not only accurate finger positioning but also the ability to adapt forces and contacts in real time. Classical motion planning struggles to scale to the dimensionality of two dexterous hands, while hand-coded controllers are brittle to variation in object placement or operator intent.

Reinforcement learning (RL) offers a principled alternative, and human demonstrations reduce the exploration burden by turning the problem into trajectory tracking [1]–[3]. Recent work shows that RL policies guided by hand-object demonstrations achieve high-fidelity bimanual manipulation in simulation [1]. For real-world deployment, however, the policy must respond to *live* human input rather than a fixed recording. At teleoperation time the incoming motion inevitably differs from any trajectory seen during training: the operator may approach from a different angle, use a different grasp, or vary timing. The central challenge is therefore *demonstration generalization* — can a single policy handle unseen trajectories of the same task?

Existing approaches address this only partially. ManipTrans [1] and DexMachina [2] each fit a separate residual policy per demonstration, and QuasiSim [4] optimizes each

trajectory independently at a cost of tens of GPU-hours per sequence. A policy trained this way memorises the specific kinematic path of that demonstration. DexTrack [3] and Dexplore [5] do train a single controller across many references, but both are driven by recorded motion-capture corpora rather than live operator input, and neither targets the bimanual residual setting. This motivates a shift in training objective: instead of fitting one demonstration, we train a single shared policy across a set of demonstrations that vary in starting configuration, approach direction and grasp style.

Our contributions are twofold:

- **A bimanual teleoperation and data-collection pipeline** integrating the Apple Vision Pro (AVP) and OptiTrack motion capture, with cross-system spatial calibration via the Umeyama algorithm [6], recording synchronized hand and object trajectories at 60 Hz and streaming them into Isaac Gym. The same hardware path serves both capture and live teleoperation without modification.
- **A single residual policy that generalizes across unseen demonstration trajectories**, demonstrated on two long-horizon bimanual tasks. In contrast to prior work [1], which requires a policy per demonstration, ours is trained jointly on demonstrations varying in starting configuration and grasp approach, and teleoperates the task on trajectories not seen during training.

II. RELATED WORK

A. Dexterous Manipulation via RL

Large-scale RL with domain randomization enables in-hand cube reorientation [7], and DeXtreme extends this to hardware through sim-to-real transfer [8]. These learn from scratch with task-specific rewards, which limits generalization to new tasks. Reference-guided RL reduces this burden by supplying a demonstration as a tracking objective [9]. Applied to dexterous hands, DexTrack trains a neural tracking controller from human references [3] and Dexplore uses reference-scoped exploration [5]. Dexplore argues that the staged decomposition — retargeting, tracking and residual correction — leaves demonstrations underused and compounds error across stages, replacing it with a single-loop optimization that treats the demonstration as soft guidance. We retain the staged design for a different reason: freezing the imitator yields a policy that runs against an arbitrary incoming target stream at control rate, which is what makes live teleoperation possible, whereas an optimization coupled to a demonstration corpus has no equivalent

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inference-time counterpart. ManipTrans [1] introduces the two-stage framework we build on, pre-training a generalist imitator on large-scale hand motion and fine-tuning a residual per demonstration; DexMachina [2] similarly trains per-demonstration policies. Both achieve strong performance on their training demonstrations but do not generalize to unseen trajectories of the same task.

B. Teleoperation and Retargeting

DexPilot formulates vision-based dexterous teleoperation around task-space fingertip objectives [10], and AnyTeleop generalizes this across embodiments [11]. OpenTeach [12] and Bunny-VisionPro [13] demonstrate low-latency teleoperation with consumer hand tracking, and Open-TeleVision [14] adds immersive visual feedback. TeleDexter [15], evaluated on single-hand manipulation, introduces the random action-masking augmentation we adopt. A common pattern across these systems is direct joint retargeting: human joint angles are mapped to robot targets via kinematic correspondences and streamed at control frequency, without a learned policy that adapts to contact forces. Recent work incorporates contact geometry into the retargeting objective itself [16]. In contrast, our approach inserts a trained RL residual between operator commands and robot actions, supplying the compliance and contact correction a retargeting-only pipeline cannot.

C. Data Collection

DexCap [17] combines a wrist-mounted SLAM camera with OptiTrack markers; OakInk-V2 [18] and ARCTIC [19] capture bimanual sequences with optical motion capture. These pipelines recover hand parameters through offline optimization-based retargeting after capture. The AVP instead provides MANO-compatible [20] keypoints directly at capture time, eliminating the post-processing stage and allowing the same pipeline to drive live teleoperation.

III. METHOD

A. Problem Formulation

A pair of robotic hands $\mathbf{d} = \{d_l, d_r\}$ must reproduce a manipulation demonstrated by human hands $\mathbf{h} = \{h_l, h_r\}$ interacting with objects $\mathbf{o} = \{o_l, o_r\}$. A demonstration provides a hand trajectory $\mathcal{T}^h = \{\tau_h^t\}_{t=1}^T$ encoding wrist 6-DoF pose, linear and angular velocity and finger joint positions, and an object trajectory $\mathcal{T}^o = \{\tau_o^t\}_{t=1}^T$. All translations are expressed relative to the wrist to reduce spatial complexity while preserving the direction of gravity. We model the task as an MDP; the action for each hand comprises target joint positions $a_q^t \in \mathbb{R}^K$ for PD control of the fingers and a 6-DoF wrench $a_w^t \in \mathbb{R}^6$ applied to the wrist.

B. Two-Stage Policy Architecture

We adopt the two-stage decomposition of ManipTrans [1] (Fig. 1). The difference lies in how we use it: instead of fitting a separate policy to each individual demonstration, we train a single shared policy across a set of demonstrations.

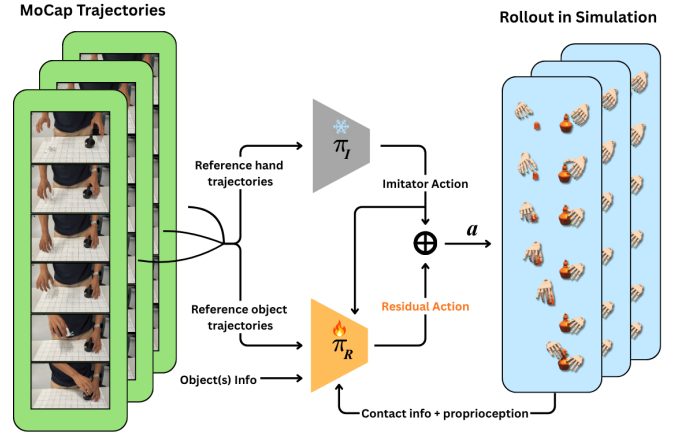


Fig. 1. System architecture. A hand-motion imitation model pre-trained on large-scale human demonstrations is frozen, and a residual policy is trained on top of it across a set of demonstrations rather than a single reference.

Stage 1 — imitator. The imitator π_I is a pre-trained generalist policy driving each hand to track the reference trajectory without object interaction. Its state is $s_I^t = \{\tau_h^t, s_{\text{prop}}^t\}$, where $s_{\text{prop}}^t = \{q_d^t, \dot{q}_d^t, w_d^t, \dot{w}_d^t\}$ are the finger joint angles and wrist pose with their velocities. We use the imitator released with ManipTrans directly and do not retrain it.

Stage 2 — residual. A residual module π_R refines the imitator output to satisfy the physical constraints of object interaction. Its state $s_R^t = s_I^t \cup s_{\text{interact}}^t$ augments the imitation state with

$$s_{\text{interact}}^t = \{\tau_o^t, p_{\delta}^t, \dot{p}_{\delta}^t, m_{\delta}, G_{\delta}, \text{BPS}(\delta), D(J_{\delta}^t, p_{\delta}^t), C^t\}, \quad (1)$$

where p_{δ}^t and $\dot{p}_{\delta}^t$ are the simulated object position (wrist frame) and velocity, m_{δ} and G_{δ} its centre of mass and gravitational force, $\text{BPS}(\delta)$ a Basis Point Set encoding [21] of the shape, $D(\cdot)$ the squared distance from each hand keypoint to the object, and C^t the per-fingertip contact force from the simulator. The correction $\Delta a_R^t \sim \pi_R(\Delta a^t | s_R^t, a_I^t, a^{t-1})$ is added element-wise, $a^t = a_I^t + \Delta a_R^t$, clipped to joint limits. Following [1] the residual is initialized with a zero-mean Gaussian and warmed up linearly, which prevents collapse. During training the imitator weights are frozen; only the residual is updated.

C. Reward

The imitator reward combines wrist tracking, finger keypoint tracking and smoothness, $r_I^t = w_{\text{wrist}} r_{\text{wrist}}^t + w_{\text{finger}} r_{\text{finger}}^t + w_{\text{smooth}} r_{\text{smooth}}^t$, with the finger term $r_{\text{finger}}^t = \sum_{f=1}^F w_f \exp(-\lambda_f \|j_{d_f}^t - j_{h_f}^t\|_2^2)$ over $F = 21$ MANO-corresponding keypoints [20], weighted towards the thumb, index and middle fingertips. For the residual stage we add two object-centric terms,

$$r_R^t = r_I^t + w_{\text{object}} r_{\text{object}}^t + w_{\text{contact}} r_{\text{contact}}^t, \quad (2)$$

where r_{object}^t penalizes positional and velocity deviation of the simulated object from its reference, and r_{contact}^t rewards fingertip force whenever the reference hand-to-object

distance falls inside a contact band (linearly ramped between 3 cm and 2 cm). The formulation deliberately avoids task-specific engineering, keeping the signal general across demonstrations.

D. Demonstration Augmentation

To expose one policy to a distribution of trajectories rather than a single path, each demonstration is augmented at load time. All spatial augmentations spin the scene about a vertical axis, so objects stay upright on the table: $p_t \mapsto R(p_t - c_t) + c_t$ with $R = R_z(\theta)$ about a per-frame pivot c_t , with the matching velocity correction. The variants differ in which bodies move and what serves as the pivot — rotating the right hand about the left hand’s object (the only augmentation that varies *inter-hand* geometry, drawn per demonstration by rejection sampling so the start position shifts by at most 5 cm), orbiting a hand about the object it already holds, and rotating an object’s orientation while the hand stays fixed, which deliberately breaks the demonstrated grasp alignment. Additive keypoint noise drawn from $\mathcal{U}(-\sigma, +\sigma)$ with $\sigma = 0.5$ cm emulates estimator error. Finally, following TeleDexter [15], a random subset of joints is periodically frozen at its previously commanded target so the policy cannot assume every degree of freedom responds on schedule; the maximum freeze duration is annealed over training.

IV. EXPERIMENTAL SETUP

A. Capture and Teleoperation Pipeline

Data collection and teleoperation use the same two systems (Fig. 2). Using the AVP SDK and VisionProTeleop [22] we collect 21 keypoints and the wrist pose per hand; OptiTrack records the object poses. The objects for the capping task are an alcohol burner and its cap, sourced from OakInk2 [18] mesh files and modified to carry OptiTrack markers; the brush task uses a $2.5 \times 20 \times 1.5$ cm prism sized after a toothbrush and a $4 \times 5 \times 4$ cm cup. Markers were placed so as to minimally affect the operator’s manipulation.

Because the two systems have independent coordinate frames, we calibrate with a tracker on the right index fingertip, moving the hand through the shared workspace and solving for the rigid map with the Umeyama algorithm [6]. The mapping is imperfect, attributable to the image-based estimation the AVP uses for finger and wrist pose and to tracker placement varying slightly between sessions; remapping error correlates positively with hand speed. At a reasonable manipulation speed of 200 mm s^{-1} to 400 mm s^{-1} the RMSE between systems is 8 mm to 12 mm.

At run time the laptop fuses both streams into a common frame, stamps each frame with a sequence number and capture time, and publishes it as `msgpack` over a ZeroMQ PUB socket at 60 Hz. The desktop subscribes and steps the policy in Isaac Gym [23] at roughly 60 Hz. Because the consumer runs below the publisher rate, the transport delivers the newest frame and discards the backlog rather than queueing it; a FIFO would let latency accumulate without bound across a session. The rendered viewer returns

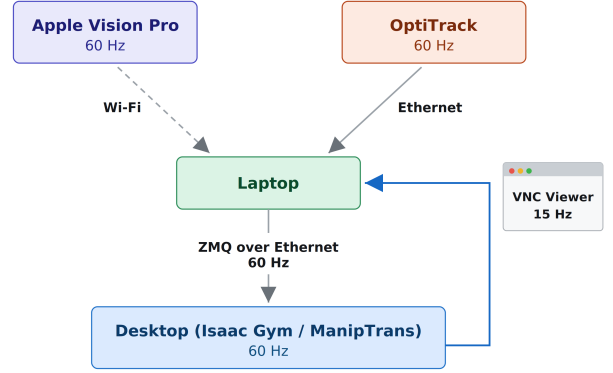


Fig. 2. Live capture and teleoperation pipeline. Annotated values are the measured rate of each stage in Hz.

to the laptop over VNC at 15 Hz, closing the loop for the operator.

B. Tasks and Demonstrations

Both tasks embody real-world scenarios (Fig. 3). In **bottle capping**, the left hand stabilizes a laboratory alcohol burner while the right hand picks up the cap and seats it; demonstrations are collected from five reach distances, spaced approximately 5 cm apart, with deliberately scattered approach angles at each distance. In **brush flipping**, the left hand grasps the brush end and lifts it, the right hand grasps the bristled end, both hands rotate the brush about its centre, the left hand releases, and the right hand places it back into the cup — a longer-horizon task with a bimanual handover. Here the brush starts at a fixed orientation and the trajectory is varied instead by lifting it to different heights and rotating it at different speeds.

C. Baselines and Metrics

We compare the full residual-imitation model against dexter-targeting [11], a classical optimization-based method, and against the frozen pre-trained imitator alone. The residual is trained with PPO [24] optimized by Adam [25] in Isaac Gym [23], on an HPC cluster with V100 and A100 GPUs; evaluation and teleoperation run on a single RTX 4090.

For offline rollouts we use three metrics following [1], averaged over successful rollouts with bimanual metrics computed per hand: object translation error E_t , object rotation error E_r , and mean per-fingertip error E_{ft} over the index and thumb, the two digits that dominate these tasks. A rollout succeeds if $E_r < 45^\circ$, $E_t < 3$ cm and $E_{ft} < 6$ cm, with both hands satisfying all conditions simultaneously. Because E_{ft} is restricted to two fingertips rather than the full set, success rates are not directly comparable with those reported in [1]. We roll out each demonstration 128 times and report the fraction that succeed. Online, task success is assessed qualitatively by visual inspection, decomposed by phase.

For generalization we partition the 50 reach demonstrations — 10 at each of five distances — into five folds by a

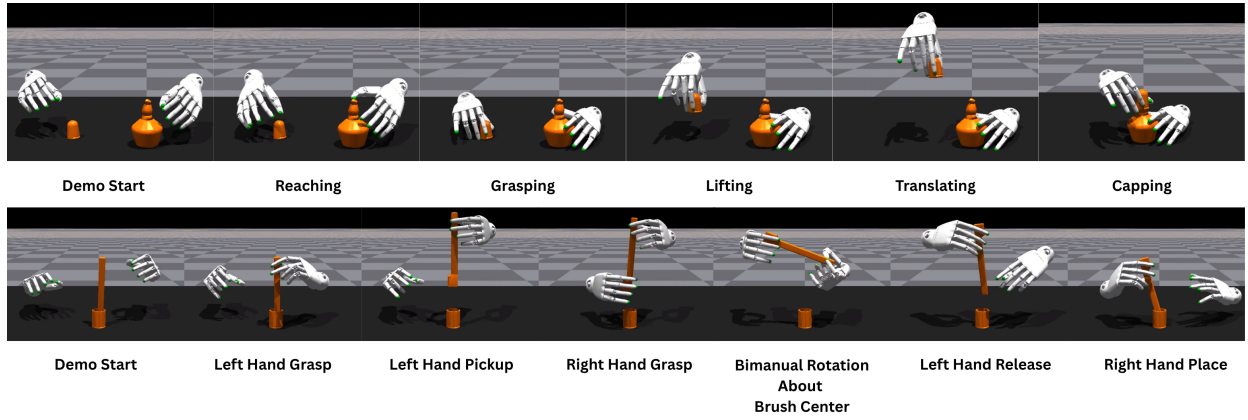


Fig. 3. Task decomposition of the bottle-capping (top) and brush-flipping (bottom) manipulation tasks. Each phase is scored independently in Tables I–II.

deterministic stratified split, dealing two per distance to each fold, so every fold holds out 10 demonstrations spanning all distances and trains on the remaining 40. Throughout, *zero-shot* denotes demonstrations held out from the policy’s training set.

V. RESULTS

A. Offline Evaluation

Table I separates the three methods by phase. All three complete the approach identically — 10/10 on reaching and stabilizing — so the comparison is decided in the contact phases.

The imitator fails at the first of them, grasping the cap in none of the 10 demonstrations. This is expected: it tracks the kinematic reference but has no mechanism to respond to contact forces. When the hand closes, the fingers reach the reference joint angles but without contact-aware correction the fingertips cannot generate sufficient grip force to lift the cap. On the brush, the same failure mode is consistent across all trials — the fingertips approach but do not close around it, and the policy proceeds open-loop as if the object were held.

The residual policy recovers this gap by conditioning on fingertip contact forces and hand-to-object distances: all 10 demonstrations grasp the cap and 8/10 lift and translate it, with 7/10 seating. On the brush it completes the full sequence in 5/10, with failures distributed across the later phases rather than concentrated at the grasp; the dominant mechanism is the handover, where the left hand fails to release cleanly after the bimanual rotation.

The dex-retargeting baseline outperforms the residual policy over the transport phases, matching it at the grasp (10/10) and exceeding it at lifting and translating (10/10 against 8/10). Solving directly for a fingertip configuration matching the operator’s hand geometry produces a repeatable, geometrically precise pinch. What it does not control is the pose of the cap *within* that pinch: without observation of fingertip contact, the thumb and index do not close simultaneously, and whichever contacts first rotates the cap between the fingertips before the grip settles. The pinch is stable — hence

TABLE I
PHASE-LEVEL SUCCESS COUNTS OVER 10 ZERO-SHOT
DEMONSTRATIONS, OFFLINE.

Task	Phase	dex-ret.	Imitator	Ours
Capping	Reach cap	10 / 10	10 / 10	10 / 10
	Grasp cap	10 / 10	0 / 10	10 / 10
	Lift cap	10 / 10	0 / 10	8 / 10
	Translate cap	10 / 10	0 / 10	8 / 10
	Seat cap	0 / 10	0 / 10	7 / 10
Brush	Lift (L)	0 / 10	0 / 10	9 / 10
	Reach (R)	0 / 10	0 / 10	8 / 10
	Rotation (Bi)	0 / 10	0 / 10	8 / 10
	Release (L)	0 / 10	0 / 10	6 / 10
	Seat (R)	0 / 10	0 / 10	5 / 10

the 10/10 transport — but stable about an orientation that differs from the one at which the cap was grasped. Nothing downstream corrects this, so the cap arrives at the bottle at an uncontrolled angle, and seating, which requires the cap axis to align with the opening within a small tolerance, succeeds in none of the 10 demonstrations. On the brush the same limitation appears earlier: the vector-based objective optimizes a different quantity than fingertip alignment, so the left hand collides with or misses the brush rather than closing around it, and the method does not progress beyond the initial reach.

B. Generalization Across Demonstrations

Fig. 4 reports the five-fold cross-validation. Held-out success varies substantially across folds and reach distances, with a mean of 0.56. Low-scoring demonstrations typically fail at the initial grasp, or form a grasp unstable enough that the cap slips from between the fingertips. Performance broadly improves as reach distance decreases, which we attribute to the hands being closer to the AVP and thus more accurately tracked. Because each fold averages only 10 demonstrations, one or two unusually hard demonstrations are enough to swing a fold mean, so individual fold numbers should not be read as evidence on their own.

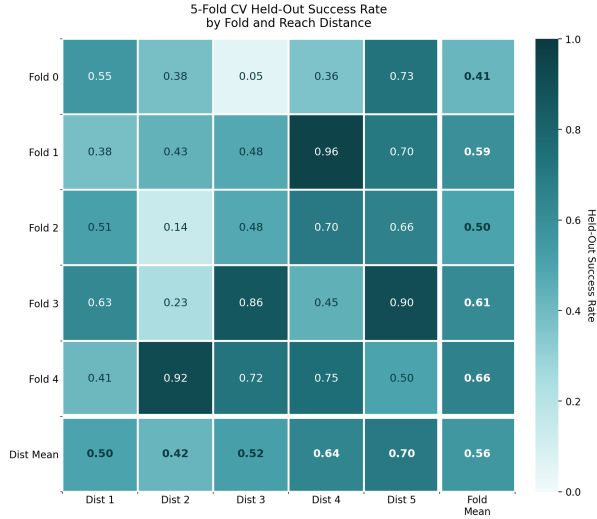


Fig. 4. Five-fold cross-validation held-out success rate by fold and reach distance. Each cell averages the two demonstrations in that (fold, distance) pair.

C. Teleoperation

Table II reports 10 consecutive live trials per task. Qualitatively, the residual model requires the least operator compensation: the grasping and capping phases, the most interaction-heavy parts of the trajectory, are assisted by the policy. It is nonetheless the only method to seat the cap at all (5/10), and completes the brush rotation in 4/10, though no method seats the brush.

Dex-retargeting is the second most difficult to operate. The cap frequently rotates during the pinch because the timing at which each fingertip contacts the surface is determined entirely by the operator’s motion; if index and thumb do not close at the same rate, the asymmetric force tilts the cap rather than gripping it stably. This forces the operator to compensate by rotating the entire wrist, making seating nearly impossible. A three-finger grasp did not help: adding a third finger raises the number of pairwise contact-timing differences from one to three, making synchronized contact harder still. Pure kinematic retargeting therefore places the full burden of contact coordination on the operator rather than the policy.

The imitation-only model is weakest: although the fingers close around the cap in all ten trials, the grasp is secure enough to lift it in only one. Its coarse positional targets bring the fingers into the vicinity of the cap but not precisely enough to seat the fingertips on its surface with enough force to retain it, so the cap slips as the hand lifts.

Two qualitative failure modes recur, both consistent with distribution shift. First, when the operator deliberately slows or pauses — for instance while visually cross-referencing the simulated hand against the real one — the frame-to-frame deltas fall outside the velocity distribution the policy was trained on. Second, the policy becomes less stable during the capping phase at the end of the episode: as the two hands converge on the burner they increasingly occlude one another

TABLE II
PHASE-LEVEL SUCCESS COUNTS OVER 10 CONSECUTIVE TELEOPERATION TRIALS.

Task	Phase	dex-ret.	Imitator	Ours
Capping	Grasp cap	7 / 10	10 / 10	9 / 10
	Lift cap	7 / 10	1 / 10	9 / 10
	Translate cap	6 / 10	0 / 10	8 / 10
	Seat cap	0 / 10	0 / 10	5 / 10
Brush	Lift (L)	7 / 10	0 / 10	10 / 10
	Reach (R)	5 / 10	0 / 10	9 / 10
	Rotation (Bi)	0 / 10	0 / 10	4 / 10
	Seat (R)	0 / 10	0 / 10	0 / 10

TABLE III
CAP-SCALING ABLATION ON THE THIN 1.6 cm CAP, SUCCESSES OUT OF 13. REACH OMITTED (13/13 THROUGHOUT).

Method	Phase	$\times 1.00$	$\times 1.15$	$\times 1.30$	$\times 1.45$
dex-ret.	Grasp	11	12	13	12
	Lift	10	12	13	11
	Translate	9	11	12	10
	Seat	0	1	1	1
Imitator	Grasp	3	8	9	8
	Lift	3	5	6	5
	Translate	3	4	5	4
	Seat	1	3	4	3
Ours	Grasp	13	13	13	12
	Lift	12	13	13	12
	Translate	11	12	12	11
	Seat	10	11	12	10

and the object, degrading the headset’s keypoint estimates at precisely the moment the tracking targets must be most accurate.

D. Ablations

Contact geometry. To isolate the role of contact force we scaled the cap both physically and in simulation (Fig. 6, Table III). A thinner physical cap (1.6 cm outer diameter against 3.0 cm) alone improves the imitator, and scaling the mesh up to $1.45\times$ improves it further: seating rises from 1/13 to 4/13. Scaling benefits the imitator most, which makes sense as it is the least accurate at kinematic retargeting. Because the imitator has no observation of the object, it closes its fingers to the same joint targets regardless of cap size, so enlarging the mesh means those fingers pinch further past the object’s original diameter; the resulting deeper interpenetration lets the simulator apply a larger contact force, supplying grip through the geometry that the imitator never learned to apply. The pattern holds for all three methods, peaking at $\times 1.30$ and declining at $\times 1.45$, where the cap becomes large enough to interfere with the grasp directly.

Augmentation. Repeating the cross-validation with the object-yaw augmentation enabled (Fig. 5) raises the mean from 0.56 to 0.60, but its effect on *invariance* is larger than on the mean: unaugmented, success broadly falls with reach distance (spread 28 points), whereas with object yaw the distance marginals flatten to 0.54–0.63 (spread 9). Rotating

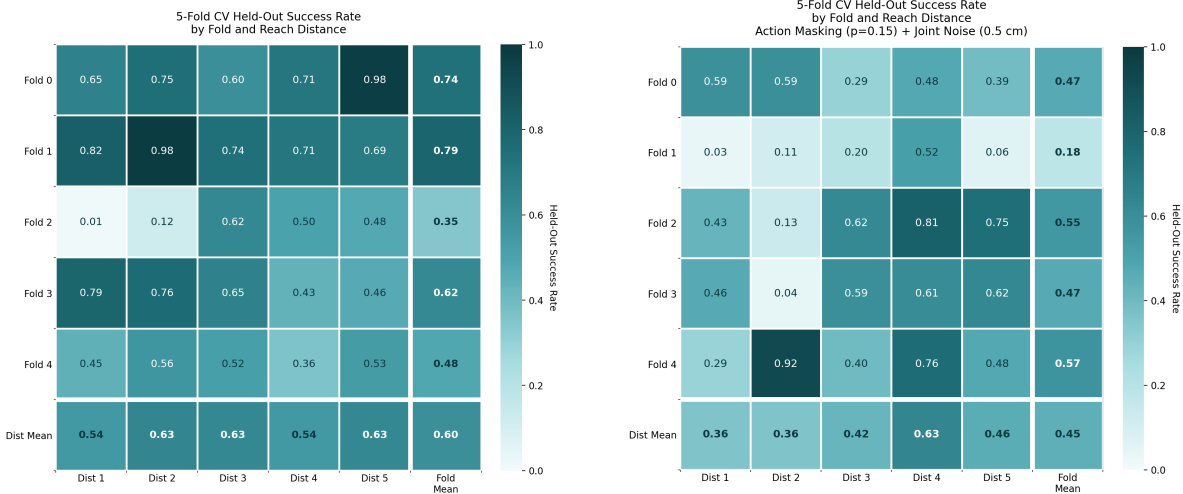


Fig. 5. Augmentation ablation under the same five-fold protocol as Fig. 4. Left: object-yaw augmentation enabled (mean 0.60), which flattens the distance marginals to 0.54–0.63. Right: action masking and keypoint noise with object yaw disabled (mean 0.45), where the distance dependence of the unaugmented baseline is unchanged.

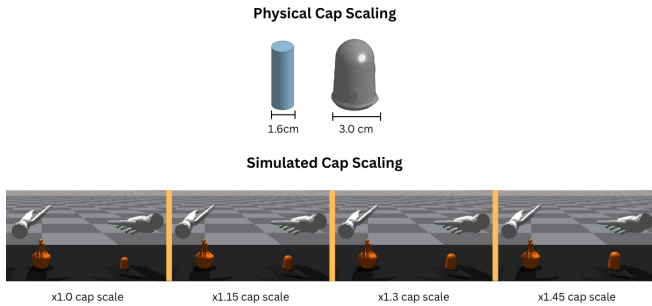


Fig. 6. Physical cap ablation (outer diameter reduced by 1.4cm) and simulated ablation by scaling the mesh.

the object about its own axis breaks the demonstrated grasp-object alignment — the burner is never replaced at the same orientation twice — so the policy cannot rely on the cap’s recorded yaw. Action masking and keypoint noise in isolation do not reproduce this benefit (mean 0.45), which is consistent with neither being designed to address reach-distance generalization: they target robustness to stale commands and to tracking-estimator error, neither of which this protocol scores.

VI. DISCUSSION & CONCLUSION

We presented a bimanual teleoperation system built on a multi-demonstration residual policy with cross-system spatial calibration, which learns contact-aware manipulation without task-specific reward engineering and operates on unseen operator trajectories without retraining. Across two long-horizon bimanual tasks it is the only method evaluated that completes the full sequence end to end in offline rollout, where both the pre-trained imitator and dex-retargeting fail at contact-critical phases. The same hardware pipeline that collects training demonstrations drives live teleoperation without modification.

Several limitations bound these results. Training one policy across many demonstrations introduces two costs: distributing a fixed number of parallel environments across demonstrations reduces the per-demonstration sample count, which can destabilize optimization; and demonstrations of differing object geometries and contact configurations may induce conflicting gradients, causing the shared policy to converge to a compromise. Sensor noise also limits learning — the AVP degrades under hand occlusion, which is exactly the regime the capping phase creates — as do inaccuracies in object meshes and their estimated poses, which caused mesh interpenetration failures in simulation despite correctly aligned real-world configurations. Finally, two aspects of the evaluation bound its scope: all demonstrations and teleoperation trials come from a single operator, so robustness to inter-operator variation in hand size, grasp preference and timing is untested; and the evaluation is carried out entirely in simulation, so the sim-to-real gap in contact dynamics, friction and actuator response is not assessed. Deployment on physical hands is the natural next step, alongside per-step target noise and speed augmentation to address the distribution shifts observed under live operation.

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